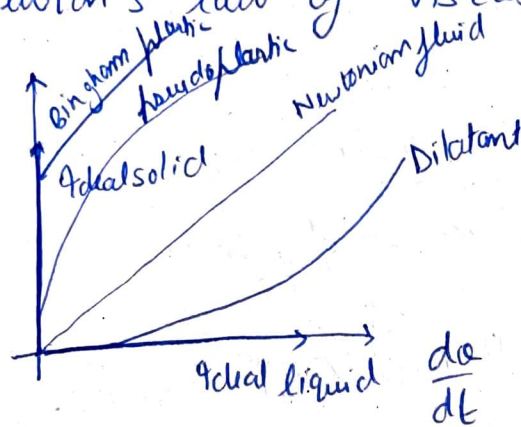


Unit - 5

→ Bernoulli Equation derivation

→ Newton's law of viscosity [Different types fluid]



$$\tau = \frac{du}{dy} \mu$$

valid for only newtonian fluids

$$F = \tau A$$

→ Pascal's Theorem



Pressure at a given point in a fluid acts equally in all direction

Unit - 4 [Thermo]

→ 1st, 2nd, 3rd law

→ calculation of COP of refrigerator

→ Otto & diesel cycle numericals & graphs

[2 constant volume]

	
$\eta_o = 1 - \frac{1}{r^{\gamma-1}}$	η_d

→ I.C engine

⇒ difference between four stroke & two stroke

⇒ do 2 numericals given in notes

Unit 3

- Classification of materials [flow chart]
 - Thermosetting and thermoset [difference]
 - Casting [gating system]
 - ⇒ use of different parts of gating system
 - Lathe Machine [diagram and parts]
 - only learn the names
 - Upmilling and downmilling [difference]
[difference on the basis of figures]
 - Soldering, Brazing
-

Lami's ✓

Variation ✓

resolution ✓

SFD & BMD ⇒ Strong Karo

Stress and strain

thermodynamics : Deals with energy interactions and its effects on system and surrounding.

⇒ System - It is a fixed mass or a region in space where our study is focussed.

⇒ Surrounding - Everything other than system is surrounding.

⇒ Boundary - between system & surrounding

X — X — X — X

Type of System	Energy trans.	Mass trans.	Example
1) Closed	✓	X	Piston cylinder without valves
2) Open	✓	✓	Piston cylinder with valves
3) Isolated	X	X	universe

Properties of system

- 1) Intensive ⇒ Independent on mass of system
example - Temperature, specific volume, pressure, density, Mol. wt, hardness.
- 2) Extensive ⇒ Dependent on mass of system
example - Entropy, Volume, energy.

Point and path function

Properties

- 1) Point function
- 2) Independent of past history
- 3) Exact differential
 $\int dT = T_2 - T_1$

work, heat

- 1) Path function
- 2) Dependent on path history
- 3) Inexact differential

Law of Thermodynamics

- Zeroth Law = It deals with thermal equilibrium. *Meaning*
- First law = Conservation of energy.
- Second law = feasibility of a process
- Third law = Absolute zero temperature

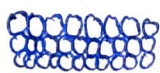
#

Macroscopic

- 1) Individual molecular behaviour taken into consideration
- 2) Applied when matter becomes rare

3) ~~mean free path~~ ^{space exploration} ~~dimension~~

Continuum: In this approach, the matter is treated to continuously distributed with no void present



no voids

Microscopic

- 1) Average molecular behaviour is considered.
- 2) Applied in continuum concept.

3) ~~mean free path~~ ^{mean free path} ~~theory~~ ^{dimension}

Mean Free Path - It is the average distance travelled by a molecule between 2 successive collisions.
Mean free path $\ll$ system dimension.

$$P + f = C + n$$

of Thermodynamic Processes

isobaric Process $\rightarrow$ In this process, the pressure remains constant. i.e. $\Delta P = 0$

2) Isochoric Process $\rightarrow \Delta V = 0$

3) Isothermal Process $\rightarrow \Delta T = 0$

Throttling Process - A process in which there is no change in enthalpy, no work is done and is adiabatic.

Zeroth Law

Acc to definition

if A is in thermal equilibrium with body B and B with C, then A is in thermal equilibrium with C.



This law is used for validation of temperature measurement

$$\frac{C}{100} = \frac{F - 32}{180} = \frac{K + 273}{100}$$

Point function

- 1) It tells us about the state of a process i.e. the P, V & T at an instant.
- 2) These are exact differentials
- 3) Eg $\rightarrow$ Pressure, volume, temperature

Path function

- 1) This depends on the path of process and cannot be denoted by a point
- 2) They are inexact differentials
- 3) Eg - Work, Energy

Energy transfer by Heat

Heat in the form of energy is transferred from a body at higher temperature to a body at lower temperature.

Heat addition $\rightarrow +Q$

Heat rejection $\rightarrow -Q$

Unit - Joules (J)

Energy transfer by Work

$$W = \int F \cdot ds$$

When work is done by the system $\rightarrow +W$

When work is done on the system $\rightarrow -W$

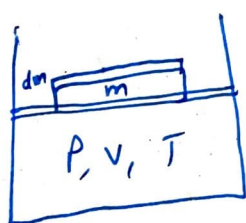
Thermodynamic Equilibrium

$\Rightarrow$ Mechanical equilibrium $\Rightarrow \sum F_i = 0$

$\Rightarrow$ Thermal equilibrium $\Rightarrow$ no change in temp

$\Rightarrow$ Chemical equilibrium $\Rightarrow$ no chemical rxn.

Quasi static process

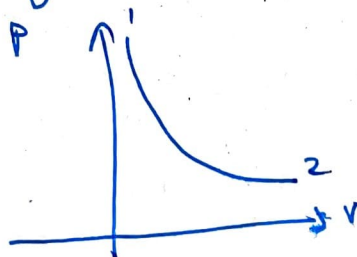


$\Rightarrow$ Weights are removed by taking infinitely small mass and then the conditions such as Pressure & Volume are plotted.

Conditions

- 1) No friction
- 2) No spontaneous changes
- 3) Heat transfer is through infinitely small dT .

$$W = \int_1^2 P \cdot dv$$



Internal Energy (U)

$U = U(T)$ we measure ΔU

Enthalpy

It is the total energy of the system

$H = U + PV$

$\gamma = \frac{C_p}{C_v}$ $\gamma = 1.66 \Rightarrow$ Monatomic
 $1.40 \Rightarrow$ Diatomic
 $1.33 \Rightarrow$ Triatomic

First Law of Thermodynamics

It is based on the law of conservation of energy.

$Q = \Delta U + W$

first law for cyclic process

$\oint dW = \oint dQ \Rightarrow$ net work = net heat transfer

Work done

1) Isothermal $\Rightarrow W = nRT \ln \frac{V_2}{V_1}$

2) Isobaric $\Rightarrow W = P(V_2 - V_1)$ or simply $\int_{V_1}^{V_2} P \cdot dV$

3) Polytropic $\Rightarrow W = \frac{n}{n-1} P_1 V_1 - P_2 V_2$

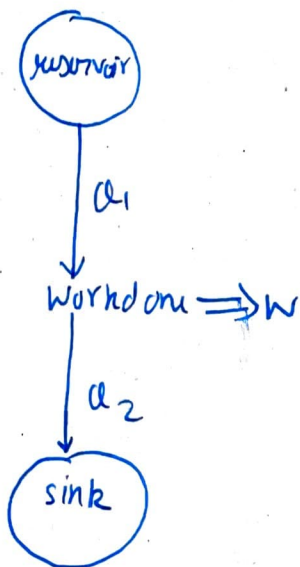
Second Law of Thermodynamics

It is mainly based on Kelvin-Planck statement and Clausius statement.

Kelvin-Planck Statement - There is no heat engine which can produce net work in a complete cycle by heat exchange at fixed temperature.

Clausius Statement - It is impossible for a device operating in a cycle to produce ~~another~~ effect other than heat transfer from hot body to cold body.

Heat engine



$$\Rightarrow \eta = \frac{\text{useful work}}{\text{heat supplied}} = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1}$$

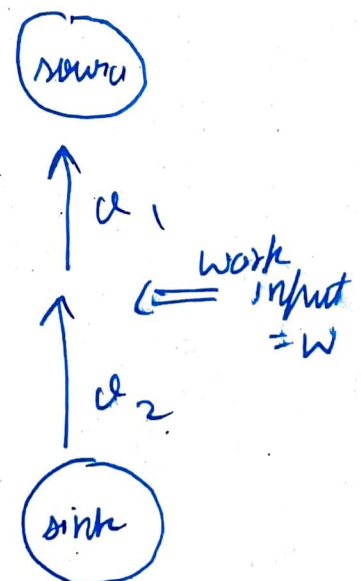
$$\boxed{\eta = 1 - \frac{Q_2}{Q_1}}$$

$$\boxed{\eta_c = 1 - \frac{T_2}{T_1}}$$

Coefficient of performance

$$COP_{ref} = \frac{\text{heat removed}}{\text{work input}}$$

$$\Rightarrow \boxed{\frac{Q_2}{Q_1 - Q_2} = \frac{T_2}{T_1 - T_2}}$$



Combustion Engine [IC-Engine]

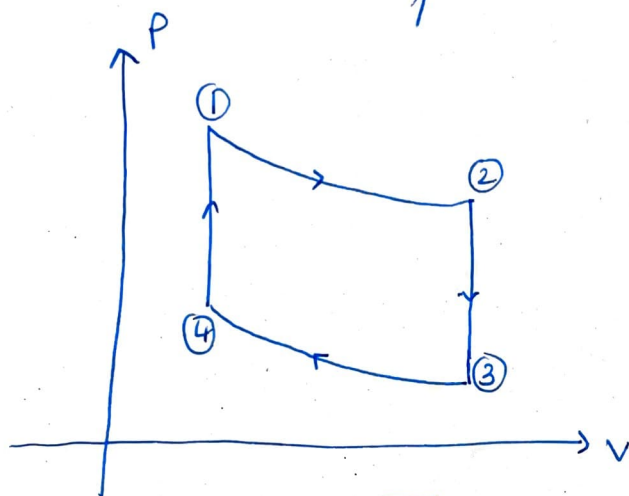
Device which converts linear motion into rotary motion by burning a fuel.

⇒ Petrol engine - Spark ignition engine (SI-engine)

Diesel engine - Compression ignition engine (CI-engine)

Otto cycle

② constant volume process
② isentropic process



1→2 [isentropic expansion]

2→3 [const-volume heat rejected]

3→4 [isentropic compression]

4→1 [const-volume heat transfer]

$$\eta_{\text{otto}} = 1 - \frac{1}{r^{r-1}}$$

$$r = \frac{V_2}{V_1} \text{ or } \frac{V_3}{V_4}$$

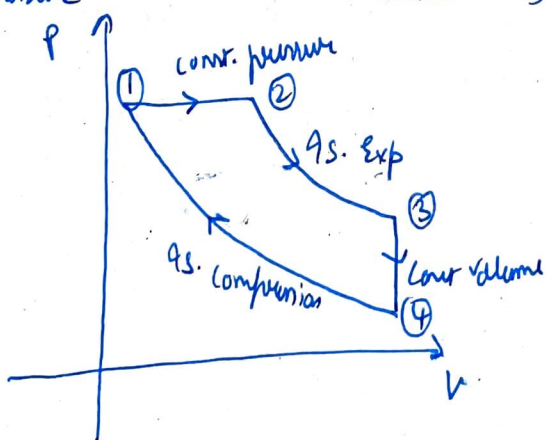
Diesel cycle

constant pressure

constant volume

is. comp

isentropic expansion



$$r = \frac{V_4}{V_1}$$

$$r = \frac{V_2}{V_1}$$

$$\eta_{\text{diesel}} = 1 - \frac{1}{r^{r-1}} \left(\frac{r^{\gamma} - 1}{\gamma - 1} \right)$$

stroke

4-stroke engine - { suction
compression
expansion
exhaust

crankshaft

acts as a flywheel and provides inertia

valve

↳ Intake - allows flowing of fuels
↳ Exhaust - removes burnt gases

Cylinder - burn fuel

Governer - controls speed of engine

Cam - control movement of valves

Carburetor - mixes fuel & air

Brake Power = $\boxed{2\pi NT = TW}$

Mechanical efficiency = $\frac{\text{output power (BP)}}{\text{input power (IP)}} \times 100$

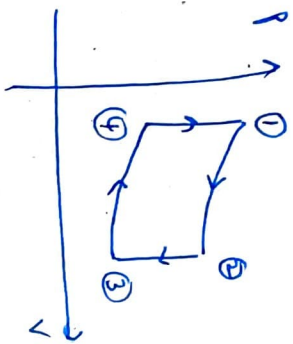
4 stroke engine

- 1) Cycle is completed in 4 strokes
- 2) Volumetric efficiency is high
- 3) Thermal efficiency is high
- 4) Flywheel is heavier

2 stroke engine

- 1) Cycle is completed in 2 strokes
- 2) Volumetric efficiency is low
- 3) Thermal efficiency is low
- 4) ~~bottom~~ flywheel is lighter

Otto cycle



1 → 2 ⇒ isentropic
expansion

2 → 3 ⇒ heat rejection

3 → 4 ⇒ isentropic
compression

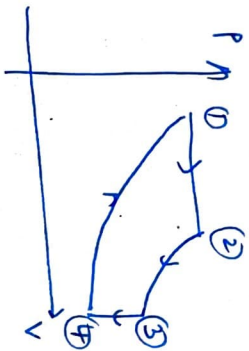
4 → 1 ⇒ heat gain

$$\mu = \frac{V_2}{V_1} = \text{expansion ratio}$$

$$\mu = \frac{V_4}{V_3} = \text{compression ratio}$$

$$\eta = 1 - \frac{1}{r^{\gamma-1}}$$

Diesel cycle



1 → 2 ⇒ isentropic comp.

2 → 3 ⇒ isobaric exp.

3 → 4 ⇒ heat rejection

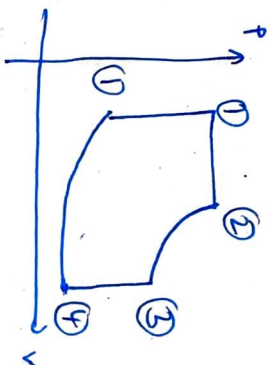
4 → 1 ⇒ isentropic comp.

$$\mu = \frac{V_4}{V_1} = \text{comp. ratio}$$

$$\rho = \frac{V_2}{V_1} = \text{cut-off ratio}$$

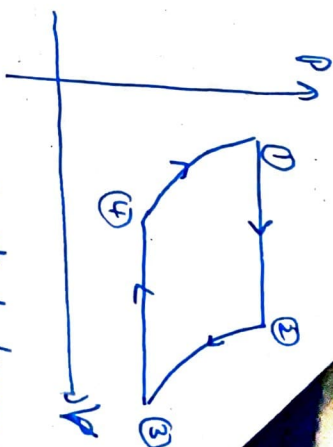
$$\eta = 1 - \left(\frac{1}{r^{\gamma-1}} \right) \left(\frac{\rho^{\gamma}-1}{\rho-1} \right)$$

Dual cycle



$$\eta = 1 - \left(\frac{1}{r^{\gamma-1}} \right) \frac{\alpha \rho^{\gamma} - 1}{\gamma(\rho-1) + 1}$$

Brayton cycle



constant pressure
1 → 2 ⇒ isentropic compression

2 → 3 ⇒ isobaric compression

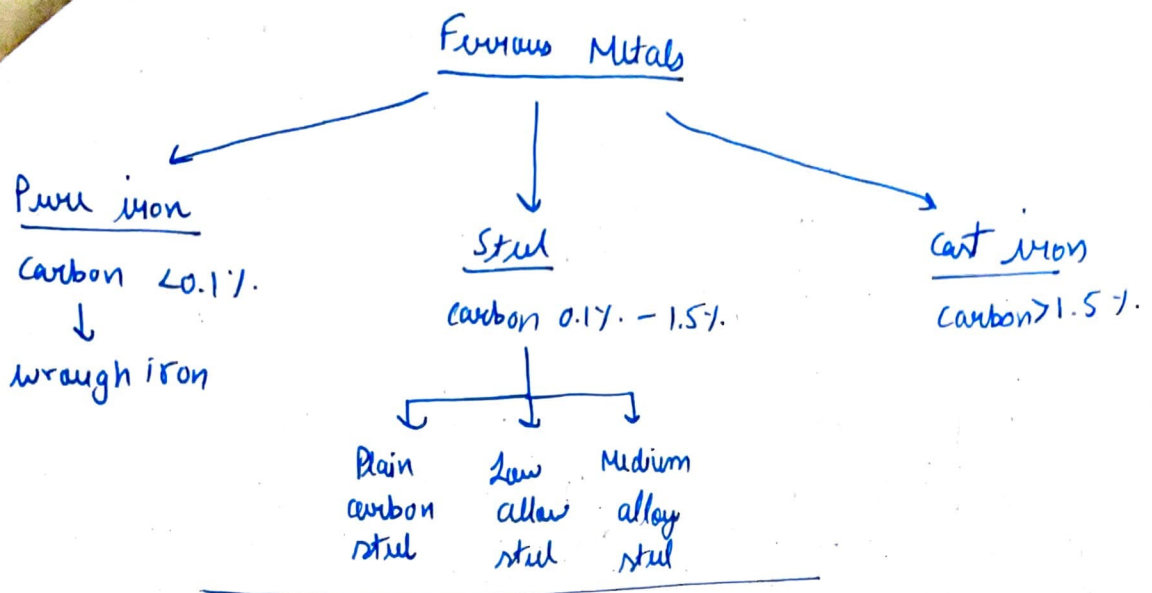
3 → 4 ⇒ isentropic expansion

4 → 1 ⇒ isobaric exp.

$$\eta = 1 - \left(\frac{T_4 - T_1}{T_3 - T_2} \right)$$

$$\eta = 1 - \frac{1}{\left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}}}$$

$$\eta_b = \frac{P_2}{P_1}$$



⇒ Thermosetting

- 1) Cannot be reheated or reshaped once set
- 2) Bond btw molecules is strong
- 3) Example - Araldite (Resin)

Thermoplastic

- 1) Remated and reshaped is possible
- 2) Bond between molecules is weak
- 3) Example - Polythene, Polythylene

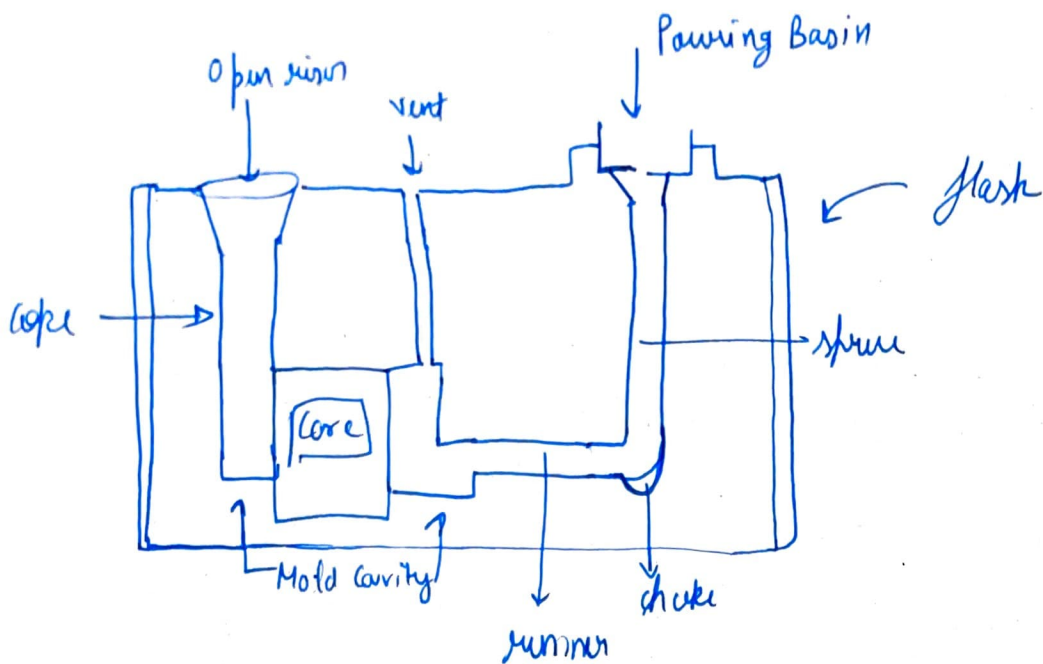
Casting

⇒ Casting is one of the oldest manufacturing process. It is the first process in making most of the products.

Basic steps

- Making mould cavity
- Material is liquified by heating
- Liquid poured in cavity
- Allowed to solidify
- Product taken out
- Trimmed and shaped

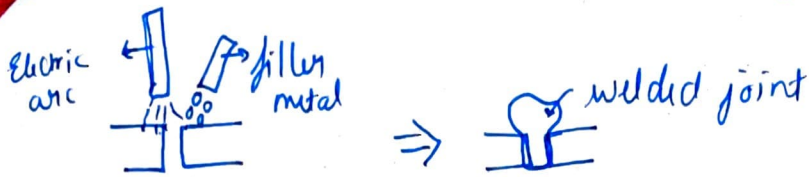
Sand Mould [Sand casting]



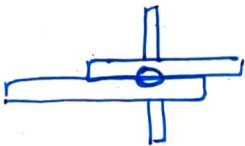
- 1) flask - metal or wooden frame without top or bottom
- 2) Pattern - It is replica of object to be made. used to make cavity
- 3) Parting line - dividing line between the two moulding flask
- 4) Core - Generally made of sand. It is used to create openings and various shaped cavities
- 5) Pouring basin - present at top of the mould and used to pour the molten metal
- 6) Spreue - Passage through molten metal flows and controls the flow of metal.
- 7) runner - pathway from spreue to the gate
- 8) Gate - channel through which molten metal is carried to cavity
- 9) Riser - contains molten metal and provides metal as casting shrinks and solidified
- 10) vent - allow escape of air

2 - provides permanent joint

MIG Welding - electric arc is used to heat base metal



2) Resistance Welding - use of resistance to produce heat between two surfaces in contact which are held in pressure



3) Diffusion welding - two parts are held together under pressure at high temperature and parts join through diffusion

4) Friction stir welding - joins through heat produced by friction

5) Ultrasonic welding - Ultrasonic frequencies are used by applying moderate pressure.

Soldering

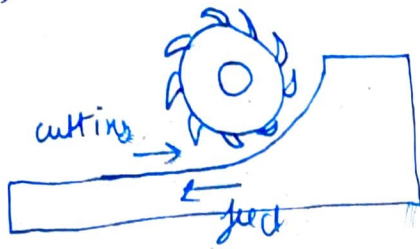
Solder is melted using heat. It is heated at around 600°F i.e. below M.P which causes it to melt and then it cools.

Brazing

In this, a filler metal is used and distributed between the surfaces in contact to be joined. Filler metals have M.P above 450°C.

Milling

1) up Milling



2) Conventional milling

3) More cutting force

4) More tool wear

1) Down milling



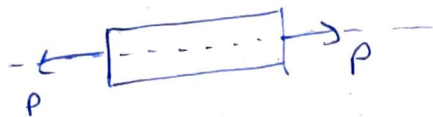
2) Climb milling

3) less cutting force

4) less tool wear

Stress

$$\sigma_T = \frac{\text{Tensile Force}}{\text{Area of cross section}}$$



2) Compression stress

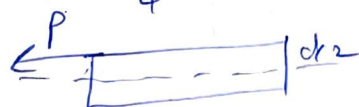
$$\sigma_c = \frac{\text{Compressive force}}{\text{Area of C.S}}$$



3) Shear stress

$$\sigma_s = \frac{\text{Shear force}}{\text{Shear area}}$$

$$= \frac{F}{\frac{\pi d^2}{4}}$$



Strain = $\frac{\Delta \text{ dimension of the object}}{\text{dimension of the object}}$

Hook's law

$$\sigma \propto e$$

$$\sigma = E e$$

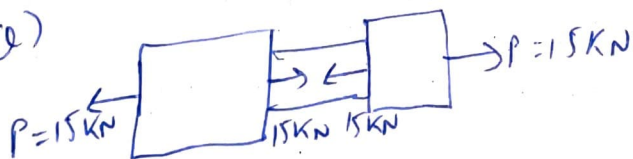
$$\Rightarrow \boxed{E = \frac{\sigma}{e}}$$

(i) Young's Modulus = $\frac{\sigma}{e} = \frac{\sigma_T}{e_t} \text{ or } \frac{\sigma_c}{e_c}$

(ii) Modulus of Rigidity = $\frac{\text{Shear stress}}{\text{Shear strain}} = \frac{\tau}{e_s}$

(iii) Bulk Modulus = $\frac{\text{Normal stress}}{\text{vol. strain}} = \frac{\sigma_n}{\frac{\Delta V}{V}}$

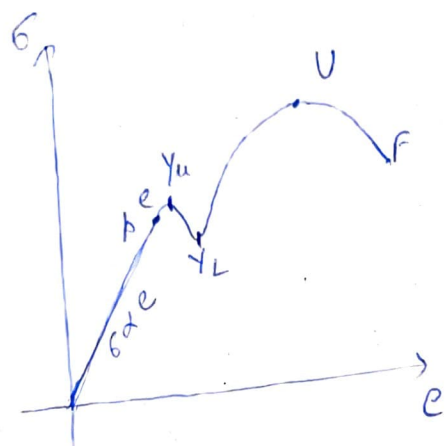
Q)



$$\Rightarrow E = \frac{\sigma}{e} \Rightarrow E = \frac{F l}{A \Delta l} \Rightarrow \Delta l = \frac{F l}{A E}$$

$$= \frac{15(0.8)}{\pi r_1^2} + \frac{15(0.5)}{\pi r_2^2} + \frac{(15)(0.2)}{\pi r_3^2}$$

Stress - Strain curve for a Ductile material



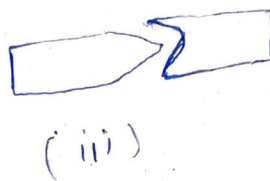
$p \Rightarrow$ limit of proportionality
 $e \Rightarrow$ elastic limit

$Y_u \Rightarrow$ upper yield point

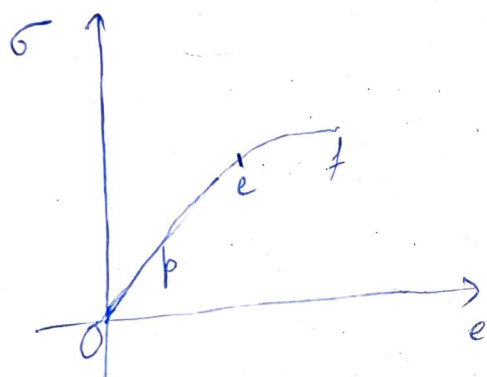
$Y_L \Rightarrow$ lower yield point

$U \Rightarrow$ ultimate stress

$F \Rightarrow$ Fracture point



Stress - Strain curve for a brittle material



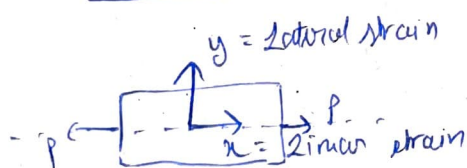
$p \Rightarrow$ limit of proportionality

$e \Rightarrow$ elastic limit

$f \Rightarrow$ fracture point



Poisson's Ratio



$$\mu = \frac{\text{Lateral strain} (\perp)}{\text{Linear strain (along force)}}$$

between the Elastic moduli

$$C = \frac{mE}{2(m+1)}$$

$C \Rightarrow$ modulus of rigidity

$E \Rightarrow$ modulus of elasticity

$K \Rightarrow$ Bulk modulus

$$(ii) K = \frac{mE}{3(m-2)}$$

$$(iii) E = \frac{9KC}{3K+C}$$

$$\boxed{m = \frac{1}{\mu}}$$

$$\text{Strain along x-axis [net]} = \frac{\sigma_x}{E} - \mu \frac{\sigma_y}{E} - \mu \frac{\sigma_z}{E}$$

$$\text{Strain along y-axis [net]} = \frac{\sigma_y}{E} - \mu \frac{\sigma_x}{E} - \mu \frac{\sigma_z}{E}$$

$$\text{Strain along z-axis [net]} = \frac{\sigma_z}{E} - \mu \frac{\sigma_x}{E} - \mu \frac{\sigma_y}{E}$$

$$\Rightarrow \Delta V = (e_x + e_y + e_z) V$$

$$= \left(\frac{\sigma_x}{E} (1-2\mu) + \frac{\sigma_z}{E} (1-2\mu) + \frac{\sigma_y}{E} (1-2\mu) \right) V$$

$$\boxed{\Delta V = \left(\frac{1-2\mu}{E} \right) (\sigma_x + \sigma_z + \sigma_y) V}$$

#1 Strain Energy

$\Rightarrow$ Resilience = Total energy stored in the material within elastic limit

$\Rightarrow$ Proof Resilience = Max. energy stored in the material with elastic limit

$\Rightarrow$ Modulus of Resilience = $\frac{\text{Proof Resilience}}{\text{Volume of body}}$

$\Rightarrow$ Strain energy, $U = \frac{\sigma^2}{2E} \times \text{Volume} = \frac{1}{2} \times \text{Stress} \times \text{Strain}$